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Title: Developing software for a writing robot

Outline of the Problem to be Solved

The outline of this project is to create a software using the C programming language that enables a writing robot to draw out text. This is done by transmitting commands from a file via a virtual RS232 serial port to the writing robot.

The goal of this project is for the robot to draw text based on font data from the 'SingleStrokeFont.txt' file. The height of the text drawn, which ranges between 4 and 10mm, will be determined by user input. The writing area has a set maximum width of 100mm, requiring that all text fits within this width without any breaks or interruptions in the words drawn.

To generate the text, the program will have to read the text content from an external file. The code written should be able to handle files of any length. Words read must be processed individually and its drawing commands be sent to the robot sequentially before reading the next word.

The G-code commands will then be generated by the software and sent to the Arduino. The G-code will instruct the robot to raise and lower the pen, as well as move the arm to specific X,Y coordinates. When drawing, each letter needs to be an offset in the X direction from the final point of the previous letter. When drawing on a new line, there will need to be a 5mm gap.

Upon completion of the drawing, the pen should return to the pen up state at position 0,0.

Key Data Items

Name	Data type	Rationale
fontData[]	int, array	This array will store the font data, read from 'SingleStrokeFont.txt'
textData[]	char, array	This array will store the text to be drawn, read from input text file
fontDataFile	FILE	The file name variable used to reference 'SingleStrokeFont.txt'
textDataFile	FILE	The file name variable used to reference the text file containing the text to be drawn
scaleFactor	float	Variable stores the calculated scale factor. Scaled by the fraction height/18 units
xPosition	int	Variable stores the current X position of the pen, also used to set horizontal offset between letters
yPosition	int	Variable stores the current Y position of the pen, also used to set vertical offset between letters
userHeight	int	Variable stores the user-defined text height, ranging between 4 to 10mm. Used for scaling.
textFontData[]	int, array	Stores font data for each character in textData[] to be drawn
gCodeCommand[]	char, array	Array stores all the G-Code commands to draw out the text

Function Declarations

*int readFontData(FILE *fontDataFile, int *fontData[])*

Parameters:

fontDataFile – font data file to be read

fontData[] – array containing font data

Return value – returns 0 if successful, 1 if failed

*int readTextData(FILE *textDataFile, char *textData[])*

Parameters:

textDataFile – text data file to be read

textData[] – array containing text data

Return value – returns 0 if successful, 1 if failed

*int calculateScaleFactor(float userHeight, float *scaleFactor)*

Parameters:

userHeight – variable containing user-defined height

scaleFactor – variable containing calculated scale factor

Return value – returns 0 if successful, 1 if failed

*int mapTextToFontData(char textData[], int fontData[], int *textFontData[])*

Parameters:

textData[] – array containing text to be drawn

fontData[] – array containing font data

textFontData[] – array containing font data of text to be drawn

Return value – returns 0 if successful, 1 if failed

*int generateGCode(int xPosition, int yPosition, int textFontData[], float scaleFactor, char *gCodeCommand[])*

Parameters:

xPosition – Variable containing current X position of the pen, used to calculate horizontal offset

yPosition – Variable containing current Y position of the pen, used to calculate vertical offset

textFontData[] – array containing font data of text to be drawn

scaleFactor – variable containing calculated scale factor

gCodeCommand[] – array containing G-Code of text to be drawn

Return value – returns 0 if successful, 1 if failed

Testing Information

Function	Test Case	Test Data	Expected Output
main	All functions success	Correct input data for all functions	0
	Any function fail	Invalid text height input	1
readFontData	Valid .txt file	SingleStrokeFont.txt	Font data loaded correctly
	Empty .txt file	EmptyFile.txt	Error message is printed, and program is terminated
readTextData	Valid .txt file	TextData.txt	Text data loaded correctly
	Empty .txt file	EmptyFile.txt	Function prints error message and terminates program
calculateScaleFactor	Valid minimum height input	userHeight = 4	Returns scale factor for the input height
	Valid maximum height input	userHeight = 10	Returns scale factor for the input height
	Invalid height input	userHeight = 3	Error message is printed, requests for valid input
	Invalid height input	userHeight = 11	Error message is printed, requests for valid input
mapTextToFontData	Valid text data values	textData[0] = { h i j }	Font data loaded correctly
	Invalid text data values	textData[0] = { 'invalid' }	Error message is printed, and program is terminated
<i>generateGCode</i>	Valid font data values	textFontData[0] = { 0 18 1 }	Correct G-Code command: gCodeCommand = S1000 G1 X0 Y18
	Invalid font data values	textFontData[0] = { 'invalid' }	Error message is printed, and program is terminated

Flowchart(s)

Attached along as PDF